

Week 3: Newton's Laws and Forces – Supplementary Notes

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These notes support the Week 3 lecture. They contain background that was moved out of the contact time. They are not needed for the lecture problems.

1 Inertial frames

Newton's first and second laws hold in an *inertial frame*: a frame in which an object with no net force moves with constant velocity. The ground is an inertial frame to a very good approximation for everyday problems. Any frame moving at constant velocity relative to an inertial frame is also inertial. A frame that accelerates (a braking car, a lift starting to move, a rotating platform) is not, and in it objects appear to accelerate with no real force acting. The “centrifugal force” felt in a turning car is such an apparent effect; in the ground frame there is no outward force.

2 The momentum form of the second law

Newton wrote the second law as $\sum \vec{F} = d\vec{p}/dt$, where $\vec{p} = m\vec{v}$ is the momentum. For constant mass, $d(m\vec{v})/dt = m d\vec{v}/dt = m\vec{a}$, giving the form used in the lecture. The momentum form is more general (for example, for a rocket whose mass changes) and is developed in the lecture on impulse and momentum.

3 Simultaneous equations: a quick refresher

Connected-object problems give two equations in two unknowns, for example

$$T - 20 = 2a, \quad 30 - T = 3a.$$

Method 1 (add): adding eliminates T : $10 = 5a$, so $a = 2.0 \text{ m/s}^2$. Substitute back: $T = 20 + 2(2.0) = 24 \text{ N}$. **Method 2 (substitute):** solve one equation for T , then put it into the other. Always check by substituting into *both* original equations.

4 Drag and terminal speed

In air or water an object experiences a drag force opposing its velocity, which grows with speed. For a falling object the net downward force $mg - F_{\text{drag}}$ falls to zero at the *terminal speed*, after which $a = 0$ and the velocity is constant. If the drag is modelled as $F_{\text{drag}} = kv^2$ (a common model at high speed), setting $mg = kv_t^2$ gives

$$v_t = \sqrt{\frac{mg}{k}}.$$

This is the first law in action: constant velocity means zero net force. It also shows why free-fall results from Week 2 apply only when air resistance is negligible.

5 The four fundamental forces

All the contact forces in this lecture (normal force, tension, friction) arise from electromagnetic interactions between atoms. In total, physics recognises four fundamental interactions:

- **Gravity:** acts between all masses; the weakest, but long-range and always attractive.
- **Electromagnetism:** acts between electric charges; long-range, attractive or repulsive; the origin of contact forces, chemistry and light.
- **Strong force:** binds quarks into protons and neutrons and holds nuclei together; very short range.
- **Weak force:** responsible for some radioactive decays; very short range.